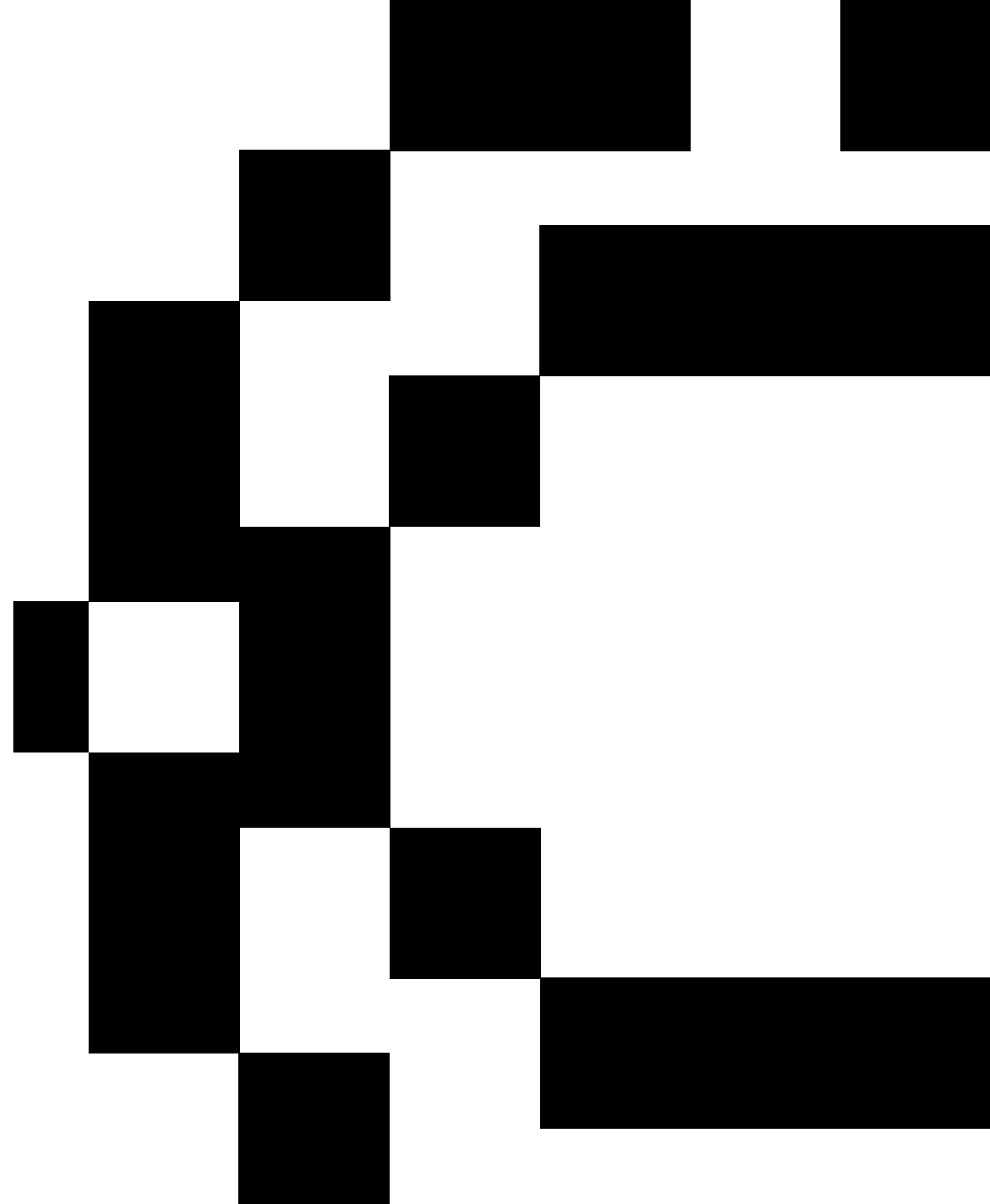
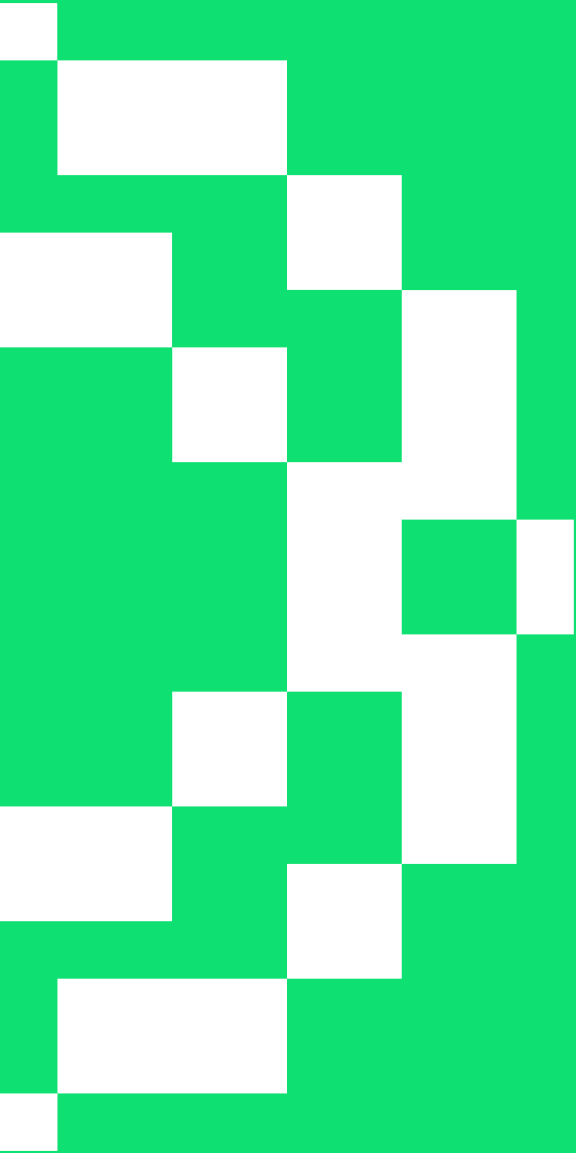


05: Model Governance

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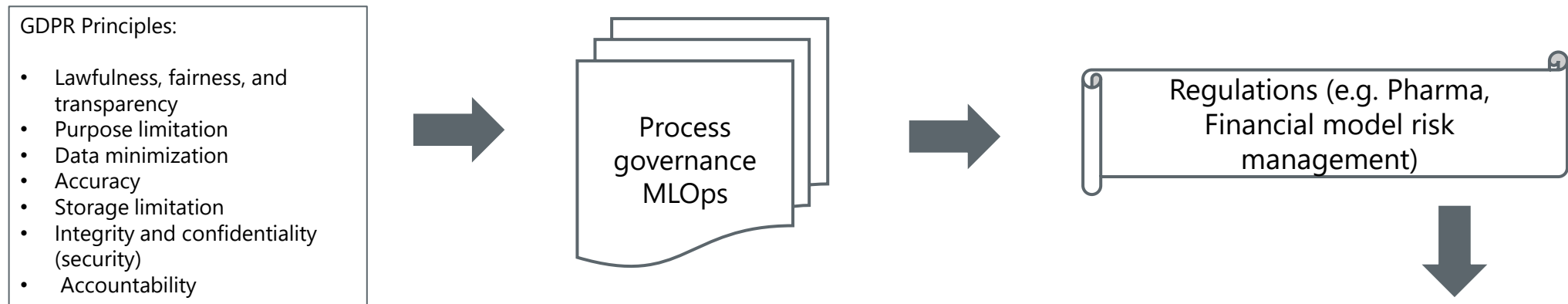




05: Model Governance

05: Model Governance

- A machine learning model is not isolated from society rules and law, and more important, the past, where the model was trained, cannot anticipate the evolving problems of the future.
- Governance is the set of controls placed on a business to ensure that it delivers on its responsibilities to all stakeholders, from shareholders and employees to the public and national governments. These responsibilities include financial, legal, and ethical obligations.



Components of Responsible AI, as an human-centered approach

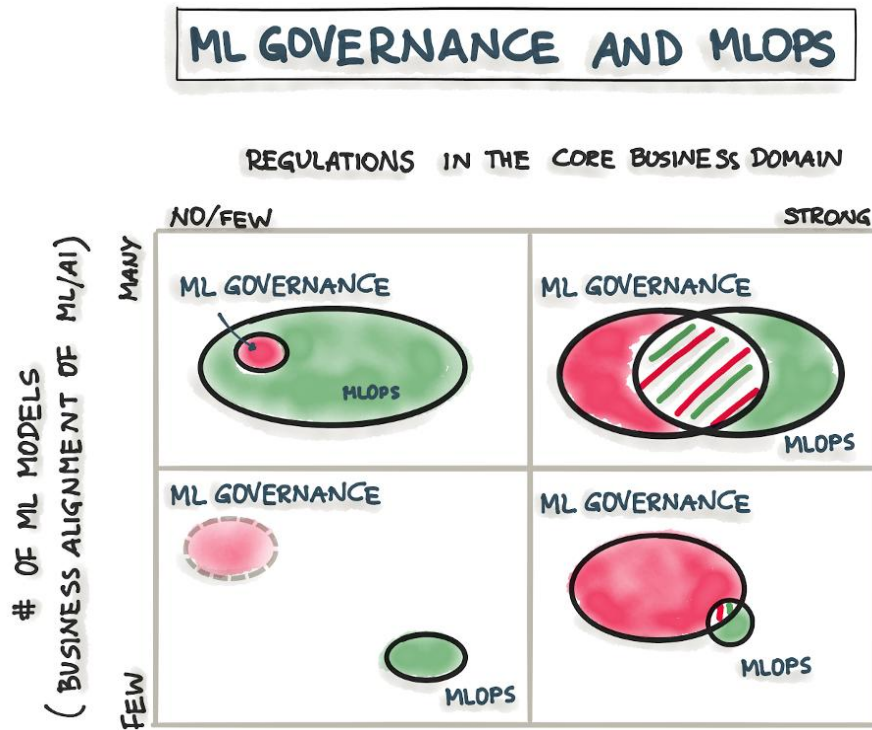
Intentionality	Accountability
Assurance that models are designed and behave in ways aligned with their purpose	Central control, management, and the ability to audit the enterprise AI effort (no shadow IT!)
Assurance that data used for AI projects comes from compliant and unbiased sources	An overall view of which teams are using what data, how, and in which models
Explainability, meaning the result of AI systems should be explainable for everyone.	Trust that data is reliable and being collected in accordance with regulation as well as a centralized understanding of which models are being used for which business process (traceability).

05: Model Governance

ML Governance is a new challenge and in the future will no be optional.

The integration of MLOps and model governance depends on two aspects:

1. **Strength of the regulations – determined by the business domain, the risk category of an ML model, and the business risk.**
2. **Number of ML models that need to be integrated into the software systems.**

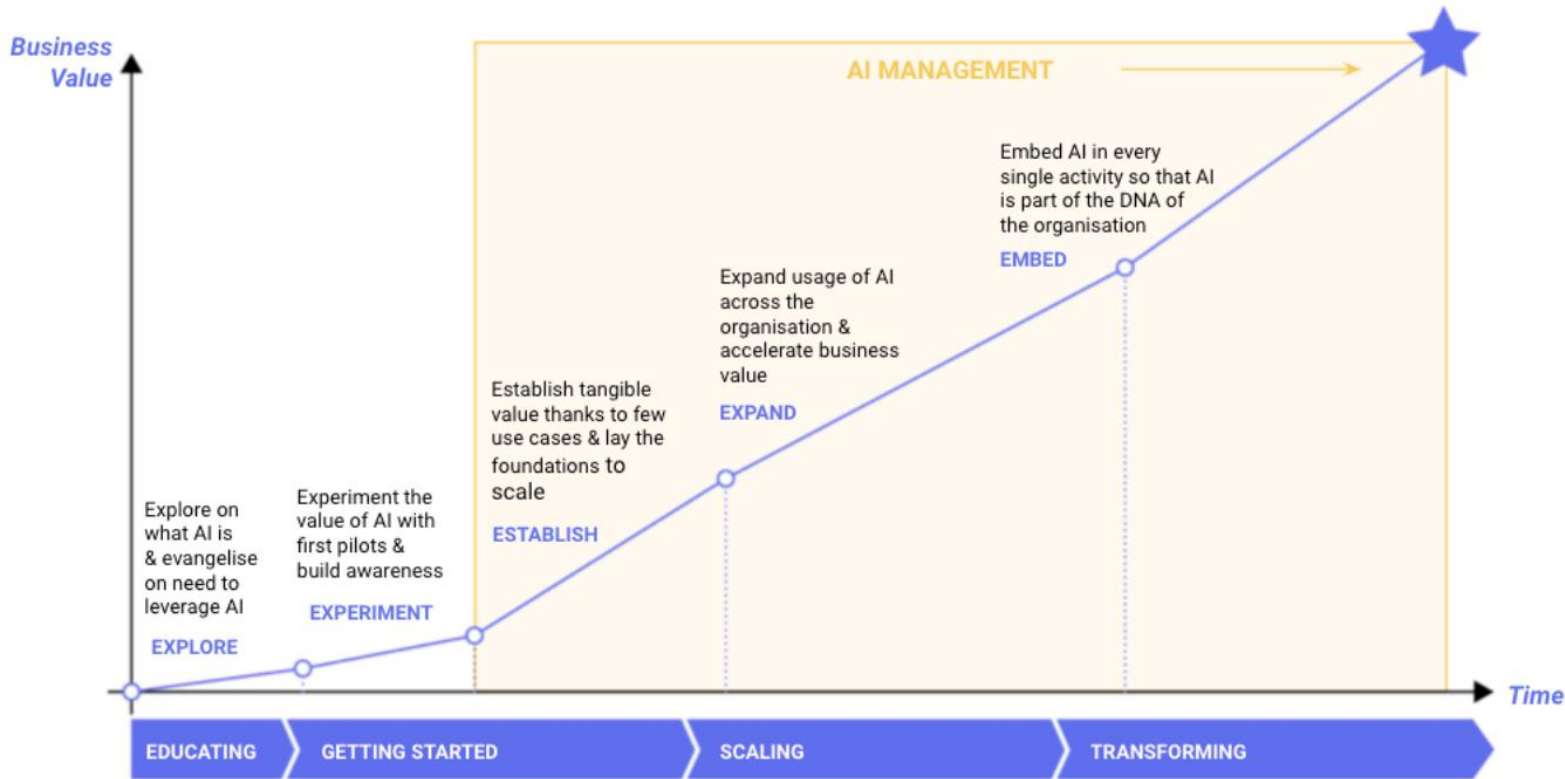


A Template for a robust governance framework of MLOps Governance:

1. Understand and classify the analytics use cases.
2. Establish an ethical position.
3. Establish responsibilities.
4. Determine governance policies.
5. Integrate policies into the MLOps process.
6. Select the tools for centralized governance management.
7. Engage and educate.
8. Monitor and refine.

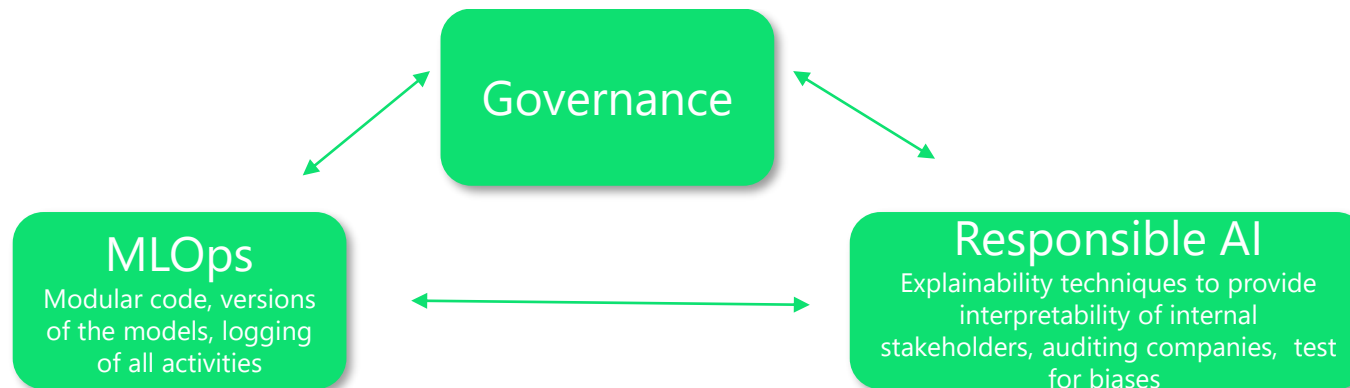
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05: Model Governance

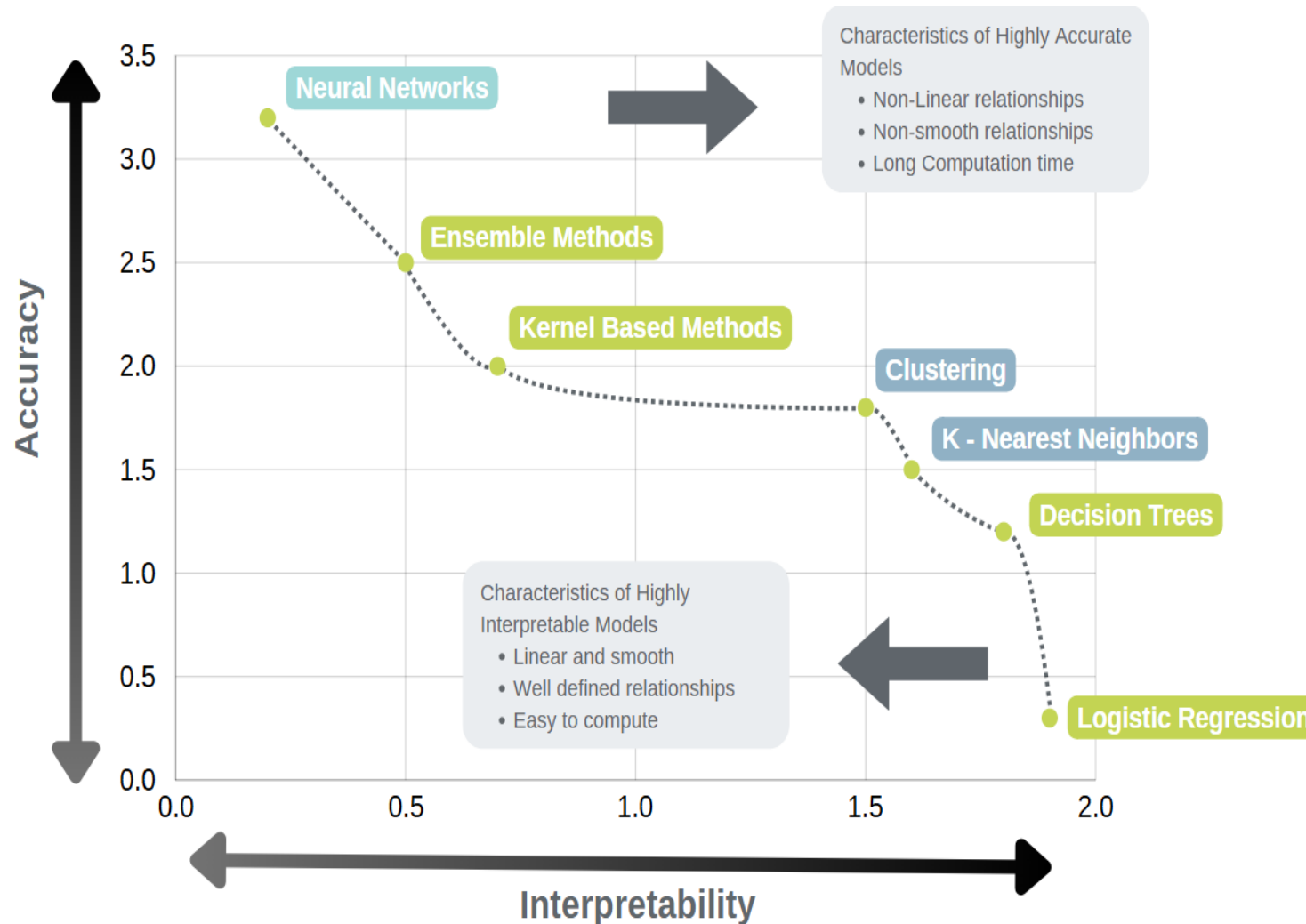


When the maturity level increases, we need to start to :

- **Document our processes;**
- **Standardize our processes (MLOps);**
- **Explain our models;**



05: Model Governance



Interpretability — If a stakeholder wants a high model transparency and wants to understand exactly why and how the model is generating predictions, **they need to observe the inner mechanics** and to interpret the model's weights and features to determine the given output.

Explainability — Explainability is how to take a model and **explain the behavior in human terms**. With complex models you cannot fully understand how and why the inner mechanics impact the prediction. We need to find methods capable of discovering the meaning between input data attributions and model outputs, which enables you to explain the nature and behavior of the model.

We are moving on a direction that these aspects will be a hard business requirement.

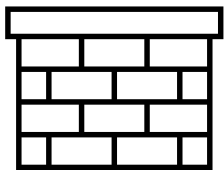
05: Model Governance

Explainability and Interpretability

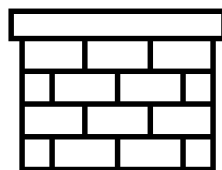
Strongest reasons to justify a full focus on Explainability and Interpretability:

- 1. Accountability:** When a model makes a wrong decision, knowing the factors that caused that decision, or who is responsible for that failure, is necessary to troubleshooting.
- 2. Trust:** In high-risk domains (like healthcare or finance), trust is critical. If you claim that your model makes better decisions and notices patterns that humans don't see, you need to have some evidence, since Domain experts will challenge your model.
- 3. Compliance:** Model explainability is critical for data scientists, auditors, and business decision-makers alike to ensure compliance with company policies, industry standards, and government regulations.
- 4. Performance:** Explainability can also improve performance, by providing the information about what to fine-tune and optimize.
- 5. Enhanced control:** Understanding the decision-making process of your models shows you unknown vulnerabilities and flaws -> easier to control and correct mistakes low-risk situations in the production environment.

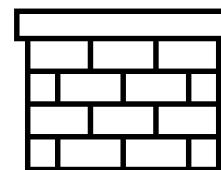
Transparency



Question



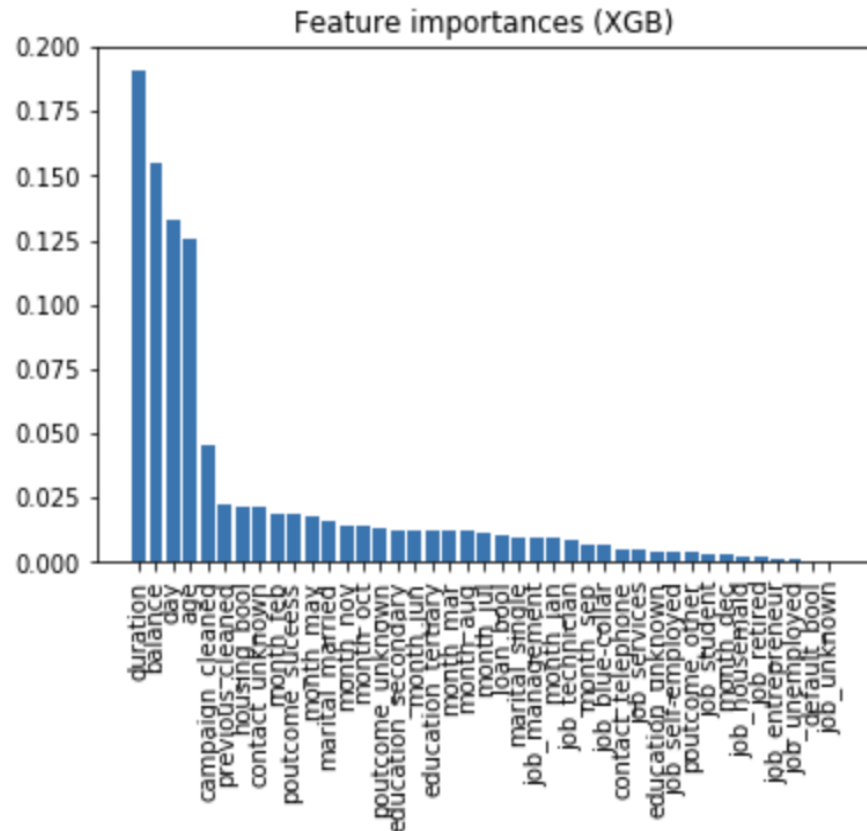
Understanding



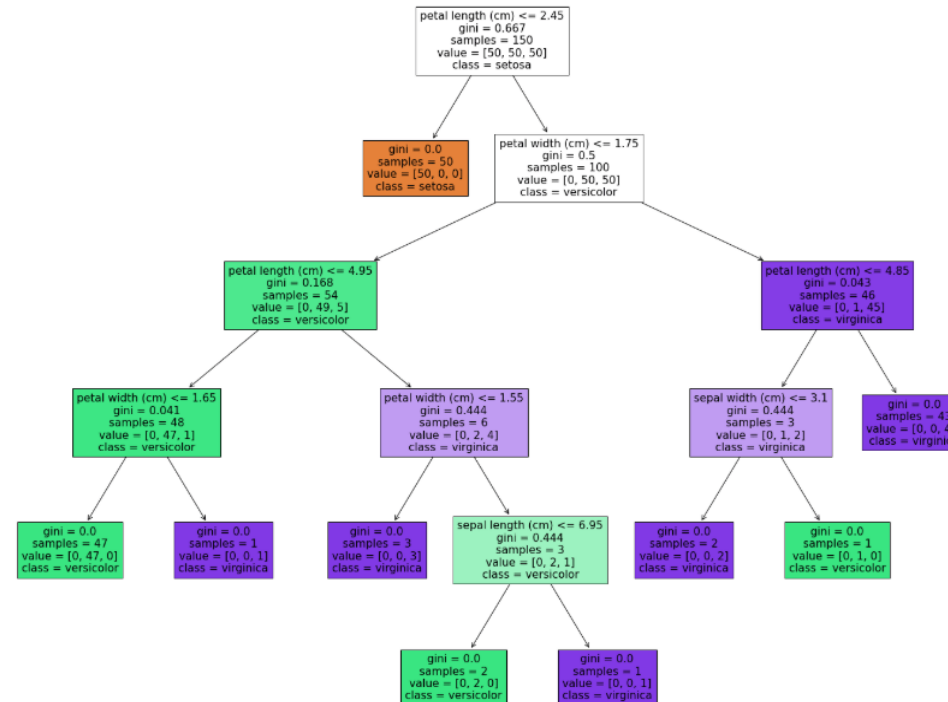
05: Model Governance

Explainability and Interpretability

- One first step to understand our model and how it works in the real world is if it is possible **look to the feature importance** (in case of tree type models check how the splits are done) and cross that information with some business metrics:



☞

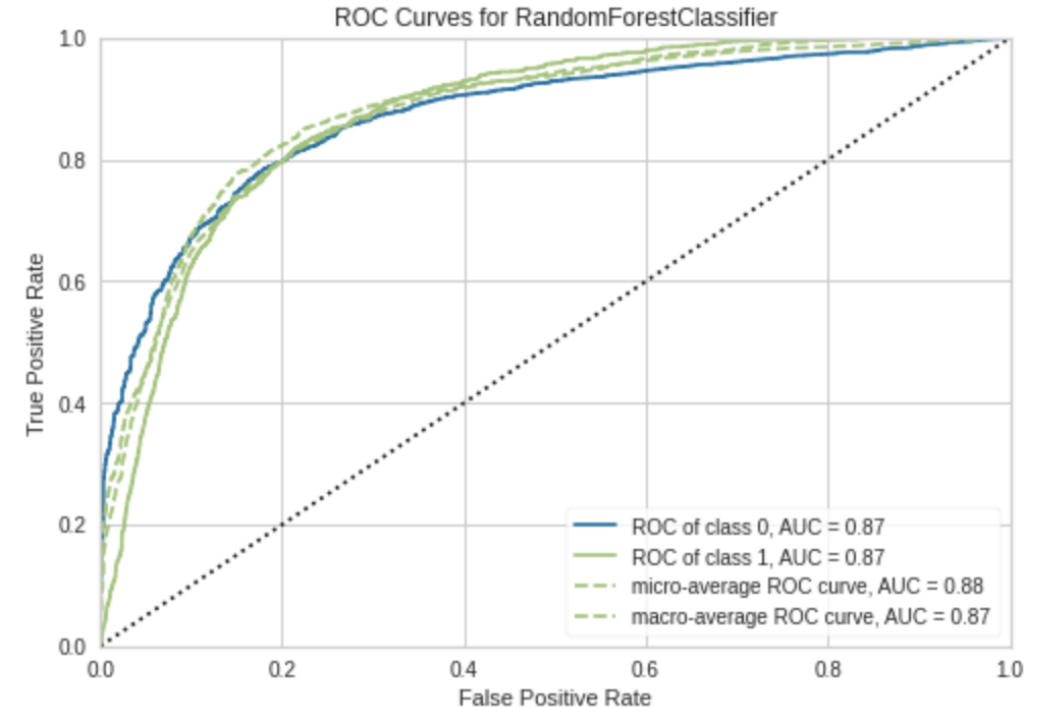
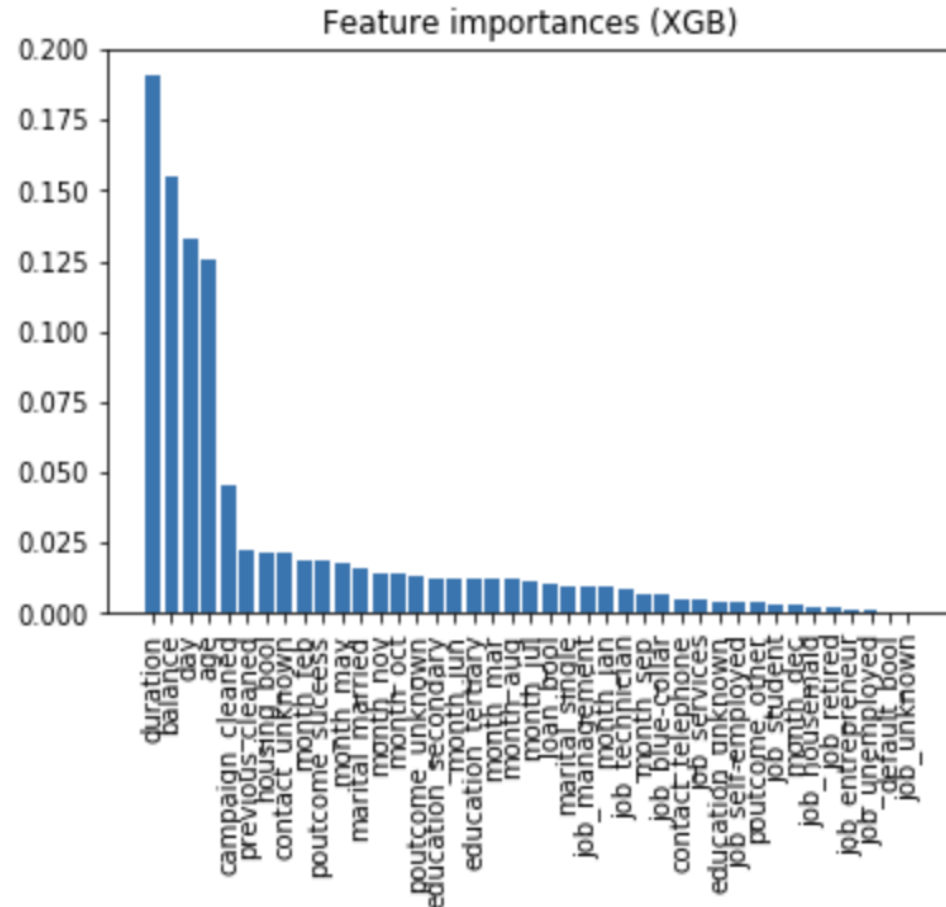


- Start by, e.g., checking the distribution of **the most relevant features when the false positive rate increases** in the ROC curve.
- The **feature importance is a good starting point to enrich our data quality tests**, more focused on what is relevant for the model.

05: Model Governance

Explainability and Interpretability

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05: Model Governance

Explainability and Interpretability

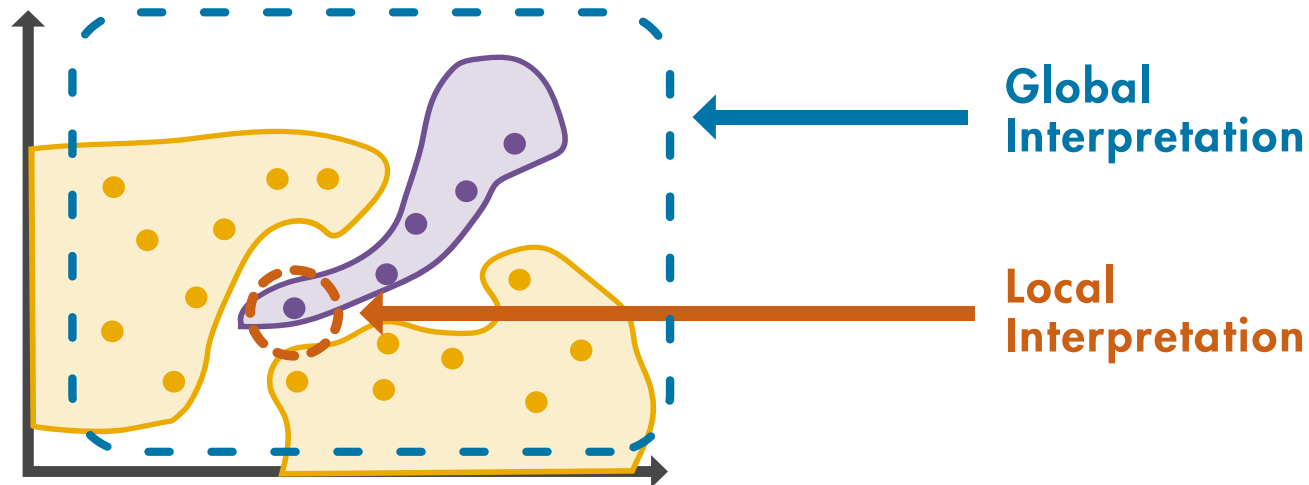
- There exists two ways of approach explainability:

Globally

This is the overall explanation of model behavior. It shows us a big picture view of the model, and how features in the data collectively affect the result.

Locally

This tells us about each instance and feature in the data individually (kind of like explaining observations seen at certain points in the model), and how features individually affect the result.



05: Model Governance

Explainability and Interpretability

In the literature you can find several methods to address some of our pain points in explaining a model and decision to an internal stakeholder, client or auditing company:

- ❑ **SHAP (SHapley Additive exPlanations)** is a framework that explains the output of any model using Shapley values, a game-theoretic approach often used for optimal credit allocation. Can be used on any black-box model (it is more efficient in tree ensembles). SHAP can be used both **globally and locally**.
- ❑ **Local interpretable model-agnostic explanations (LIME)** is a method that fits a **surrogate glass-box model around the decision space of any black-box model's prediction**. LIME explicitly tries to model the local neighbourhood of any prediction. LIME works by perturbing any individual data point and generating synthetic data which gets evaluated by the black-box system. LIME has been designed to be applied **locally**.
- ❑ **Permutation Importance**, where **feature importance** can be measured by looking at **how much the score** (accuracy, F1, R^2 ,...) **decreases when a feature is not available**. To do that one can remove a feature from the dataset, re-train the estimator and check the score. Of course, permutation importance can only be applied **globally**.
- ❑ **Partial dependence plot (short PDP or PD plot)** shows the marginal effect one or two features have on the predicted outcome of a machine learning model. A partial dependence plot can show whether the **relationship between the target and a feature** is linear, monotonic or more complex. DP assumes independence between the features and can lead to misleading interpretability-wise if the independence conditions are not met. It can only be applied **globally**.
- ❑ **Tree Surrogates** are an interpretable model that is trained to approximate the predictions of a black-box model. We can draw conclusions about the black-box model by interpreting the surrogate model. The policy trees are easily human interpretable and provide quantitative predictions of future behaviour. Can be used both **globally and locally**.
- ❑ **Explainable Boosting Machine (EBM)** is an interpretable model developed at Microsoft Research. It is a tree-based, cyclic gradient boosting Generalized Additive Model with automatic interaction detection. EBMs are often as accurate as state-of-the-art blackbox models while remaining completely interpretable. Although EBMs are often slower to train than other modern algorithms, EBMs are extremely compact and fast at prediction time. EBM can be used both **globally and locally**.

<https://christophm.github.io/interpretable-ml-book/>



05: Model Governance

Permutation Importance

05: Model Governance

Permutation Importance

We measure the importance of a feature by **calculating the increase in the model's prediction error after permuting the feature**.



A feature is important if shuffling its values increases the model error (the model relied on the feature for the prediction)

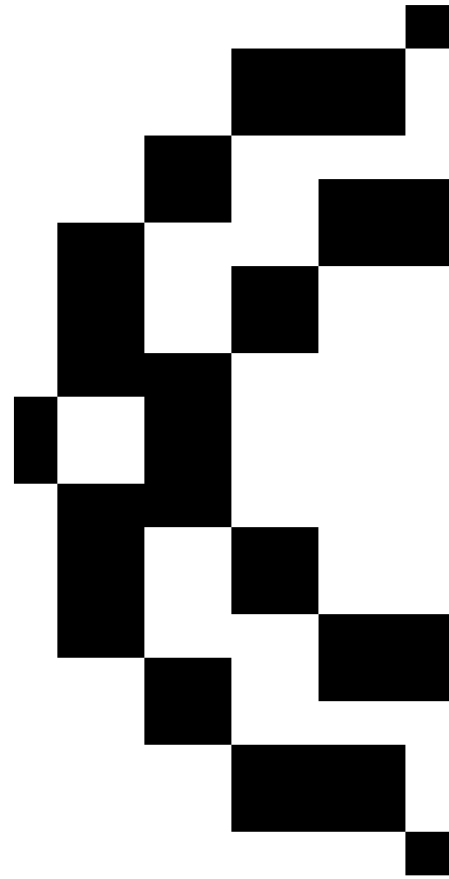


A feature is “unimportant” if shuffling its values leaves the model error unchanged, because in this case the model ignored the feature for the prediction.

Input: Trained model $\hat{f}$, feature matrix X , target vector y , error measure $L(y, \hat{f})$.

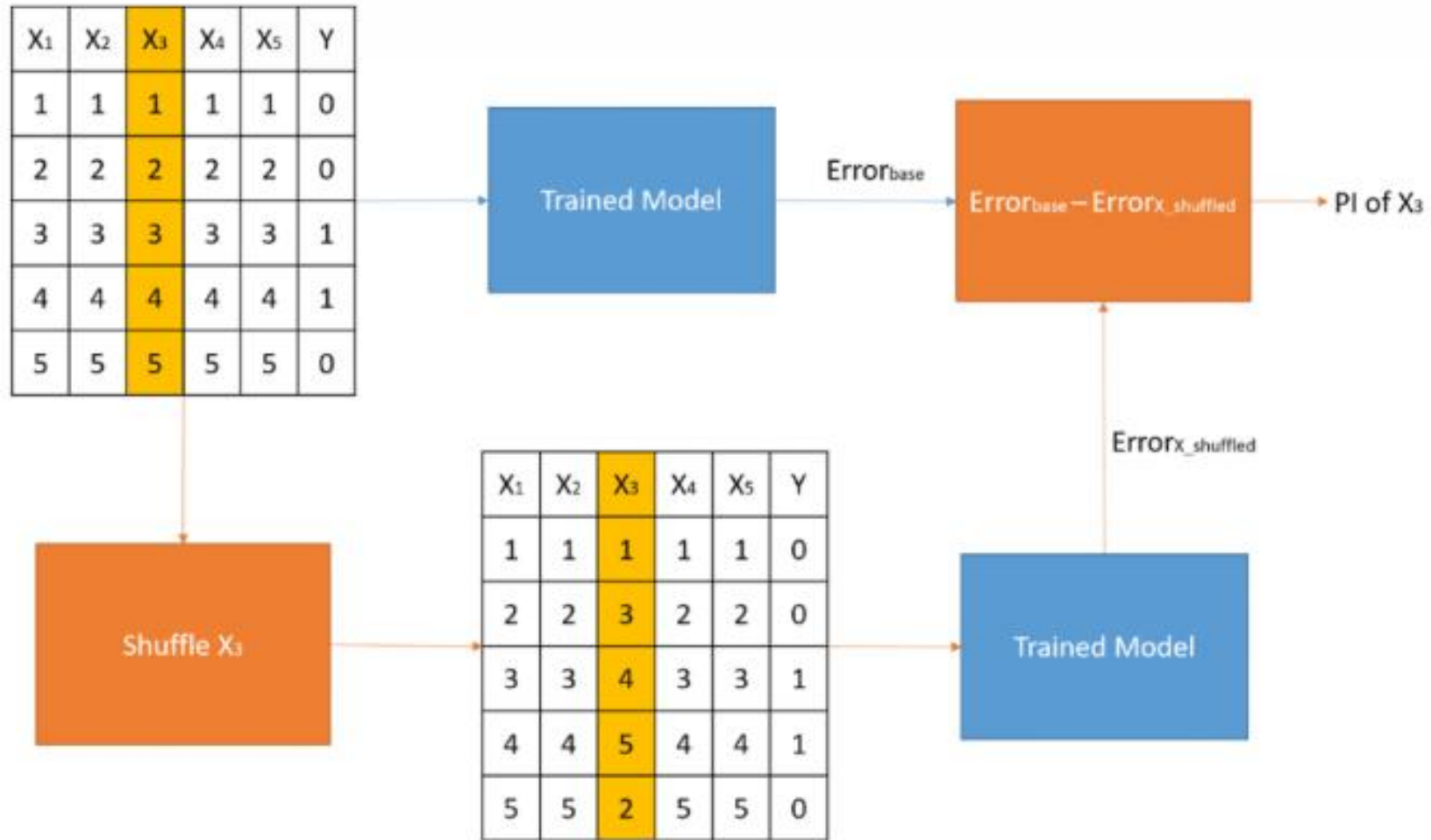
1. Estimate the original model error $e_{orig} = L(y, \hat{f}(X))$ (e.g. mean squared error)
2. For each feature $j \in \{1, \dots, p\}$ do:
 - Generate feature matrix X_{perm} by permuting feature j in the data X . This breaks the association between feature j and true outcome y .
 - Estimate error $e_{perm} = L(Y, \hat{f}(X_{perm}))$ based on the predictions of the permuted data.
 - Calculate permutation feature importance as quotient $FI_j = e_{perm}/e_{orig}$ or difference $FI_j = e_{perm} - e_{orig}$
3. Sort features by descending FI.

<https://arxiv.org/abs/1801.01489>



05: Model Governance

Permutation Importance



05: Model Governance

Permutation Importance

Advantages of the method:

- It is easy to understand and compute, as it **does not require retraining the model**.
- Can quickly provide a **global explanation of how a model makes its decisions** based on the feature set.
- The trained model captures all feature interactions -> **considers both the main feature effect and the interaction effects on model performance**.

Disadvantages of the method:

- Cannot explain the **predictions for specific data points or instances**.
- You **need access to the true outcome**.
- The **results might vary greatly**. Repeating the permutation and averaging the importance measures over repetitions stabilizes the measure but increases the time of computation.
- If the features are correlated, the permutation feature importance **can be biased by unrealistic data instances**.

One idea can be finding a solution that provides p-values for the importances

<https://academic.oup.com/bioinformatics/article/26/10/1340/193348?login=false>



05: Model Governance

SHAP

05: Model Governance

SHAP

- Method inspired on cooperative coalitional game theory. Cooperative game theory originally looked to identify **how the cooperation between groups of players (“coalitions”) contributed to the gain of the coalition overall (payout) -> how features contribute to the model’s outcome.**
- **Players cooperate in a coalition and receive a certain profit from this cooperation**, depending on their contribution to the total payout, **much as the features of a model:**
 - The sum of what everyone receives should equal the total reward;
 - If two people contributed the same value, then they should receive the same amount from the reward;
 - Who does not contribute does not receive any value;
 - If the group plays two games, an individual should win a reward from both games equal their reward from both games.

05: Model Governance

SHAP

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Subgroup	Subgroup output	Marginal contribution of A	Marginal contribution of B	Marginal contribution of C
A	4	4	.	.
B	4	.	4	.
C	4	.	.	4
A, B	9	5	5	.
A, C	10	6	.	6
B, C	11	.	7	7
A, B, C	15	4	5	6
Shapley value	.	4.5	5	5.5

Table A

The feature values enter a game in random order. All feature values participate, contributing to the prediction. The Shapley value of a feature value is the average change in the prediction that the coalition already in the game receives when the feature value joins them.

05: Model Governance

SHAP

- The Shapley value of a feature value is its contribution to the payout, weighted and summed over all possible feature value combinations:

$$\phi_j(val) = \sum_{S \subseteq \{1, \dots, p\} \setminus \{j\}} \frac{|S|! (p - |S| - 1)!}{p!} (val(S \cup \{j\}) - val(S))$$

Weighting all feature combinations

The marginal contribution of feature j

where S is a subset of the features used in the model, x is the vector of feature values of the instance to be explained and p the number of features. $val(S)$ is the prediction for feature values in set S that are marginalized over features that are not included in set S :

$$val_x(S) = \int \hat{f}(x_1, \dots, x_p) d\mathbb{P}_{x \notin S} - E_X(\hat{f}(X)) \quad val_x(S) = val_x(\{1, 3\}) = \int_{\mathbb{R}} \int_{\mathbb{R}} \hat{f}(x_1, X_2, x_3, X_4) d\mathbb{P}_{X_2 X_4} - E_X(\hat{f}(X))$$



Evaluate the prediction for the coalition S consisting of feature values x_1 and x_3

- Summing all shapley values of all features, we obtain:

$$\sum_{j=1}^p \phi_j = \hat{f}(x) - E_X(\hat{f}(X))$$

The difference between the prediction and the average prediction is **fairly distributed** among the feature values of the instance

05: Model Governance

SHAP

A regression problem where the task is to develop a machine learning model that predicts a continuous house price based on 12 features

Waterfall plots

Colour = Actual Feature Value

X-Axis = Impact on Model Prediction

The model's predicted house price for this example.

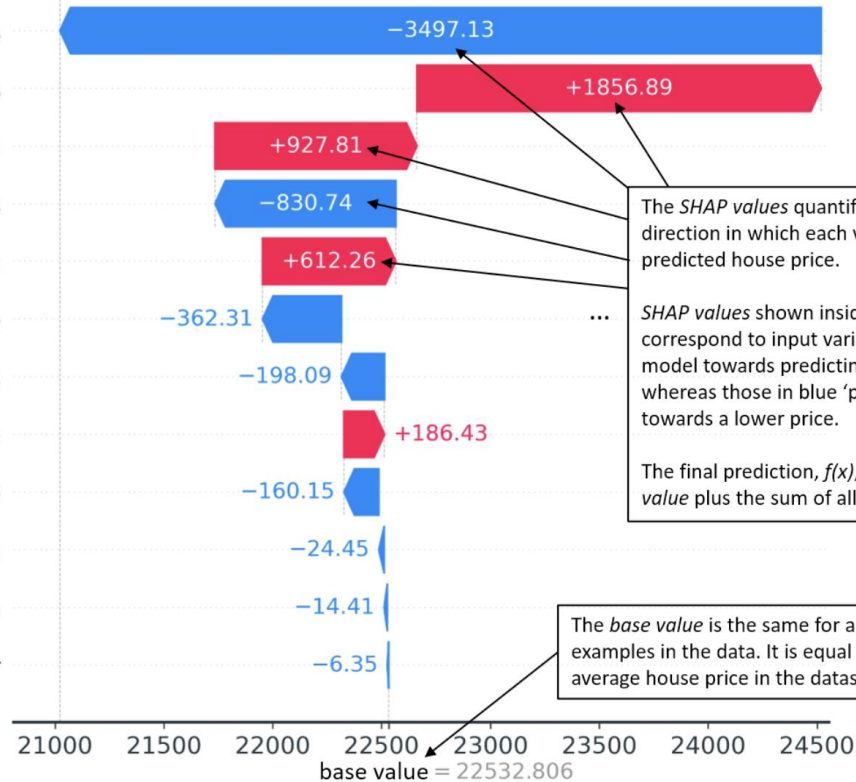
$$f(x) = 21022.57 = \text{base value} + \text{sum}(\text{SHAP values})$$

A **waterfall plot** provides a detailed breakdown of how each input variable contributes towards the predicted house price for a single instance of the data.

These are the input variables, ranked from top to bottom by how much impact they have on the model's prediction for this example from the data.

The grey numbers denote the values of the variables for this particular instance.

5.878 = number of rooms
16.2 = % working class
21.4 = % built before 1940
6.498 = remoteness
0.409 = NOX concentration
345 = tax rate
4 = connectedness
0.058 = crime rate
6.07 = % industrial zone
18.9 = pupil-teacher ratio
12.5 = % residential zone
0 = Charles River



The *SHAP values* quantify the amount and direction in which each variable impacts the predicted house price.

SHAP values shown inside red arrows correspond to input variables that 'push' the model towards predicting a higher price, whereas those in blue 'push' the model towards a lower price.

The final prediction, $f(x)$, is equal to the *base value* plus the sum of all the *SHAP values*.

The *base value* is the same for all examples in the data. It is equal to the average house price in the dataset.

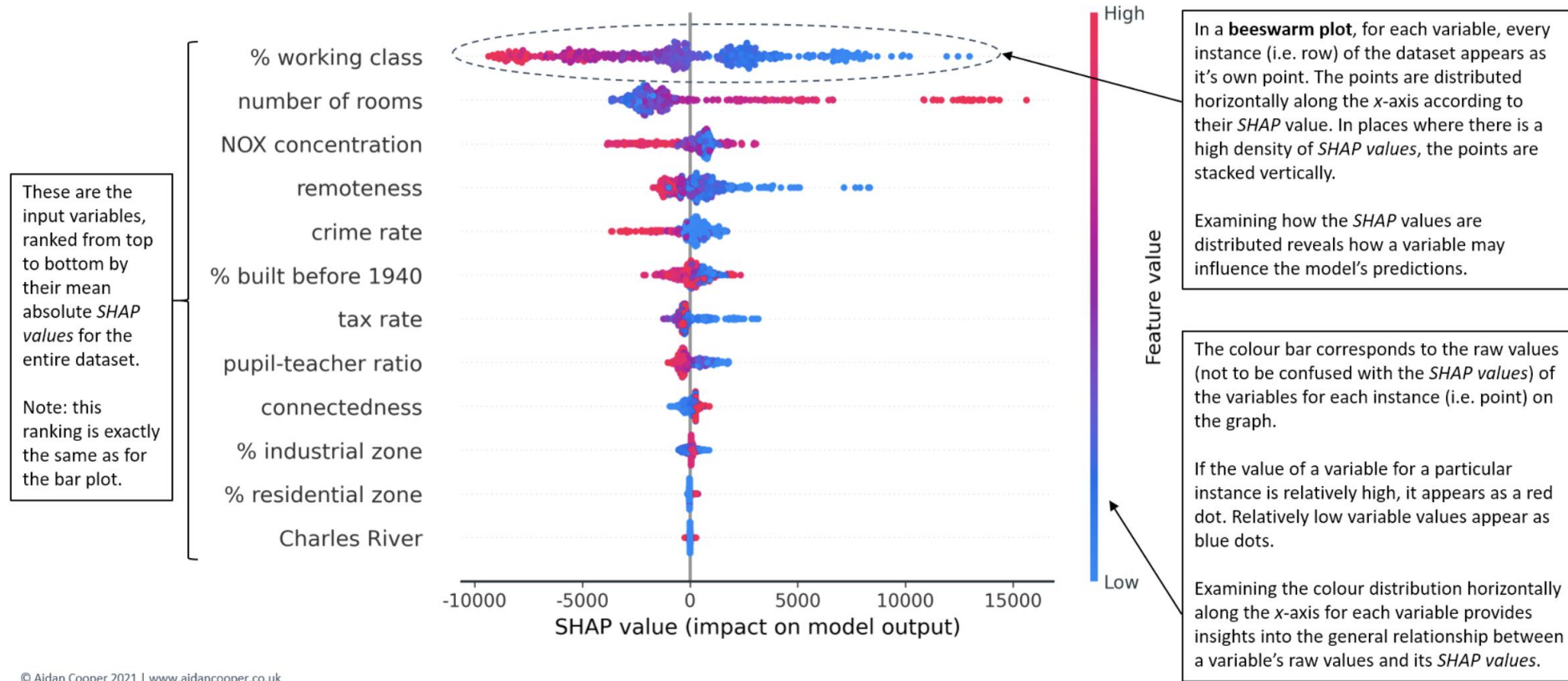
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- Shapley values calculate the importance of a feature by **comparing what a model predicts with and without the feature**.
- The order in which a model sees features can affect its predictions. The exercise is done in **every possible order**, so that the features are fairly compared.

05: Model Governance

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